

Name: \_\_\_\_\_

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Score: \_\_\_\_\_

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**Directions:** Answer in complete logical sentences. Put phones away. No calculators, etc.

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1. Suppose  $A$  is a fixed  $2 \times 2$  matrix. Show that the set  $W = \{X : AX = XA\}$  is a subspace of  $M_{2,2}$ .

(a) Suppose that  $B$  and  $C$  are matrices in the set  $W$ .

This means  $AB = BA$  and  $AC = CA$ .

Then  $A(B + C) = AB + AC = BA + CA = (B + C)A$ .

And  $A(B + C) = (B + C)A$  means that  $B + C \in W$ .

Therefore  $W$  is closed under addition.

(b) Suppose that  $B \in W$  and  $c \in \mathbb{R}$ .

The fact that  $B \in W$  means  $AB = BA$ .

Observe that  $A(cB) = c(AB) = c(BA) = (cB)A$ .

And  $A(cB) = (cB)A$  means  $cB \in W$ .

Therefore  $W$  is closed under scalar multiplication.

Parts (a) and (b) above show that  $W$  is closed under addition and scalar multiplication, so  $W$  is a subspace.